

MI HDMI API

Version 2.13

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REVISION HISTORY

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2.13	• Added Enum E_MI_HDMI_COLOR_TYPE_AUTO E_MI_HDMI_OUTPUT_MODE_AUTO	02/26/2022

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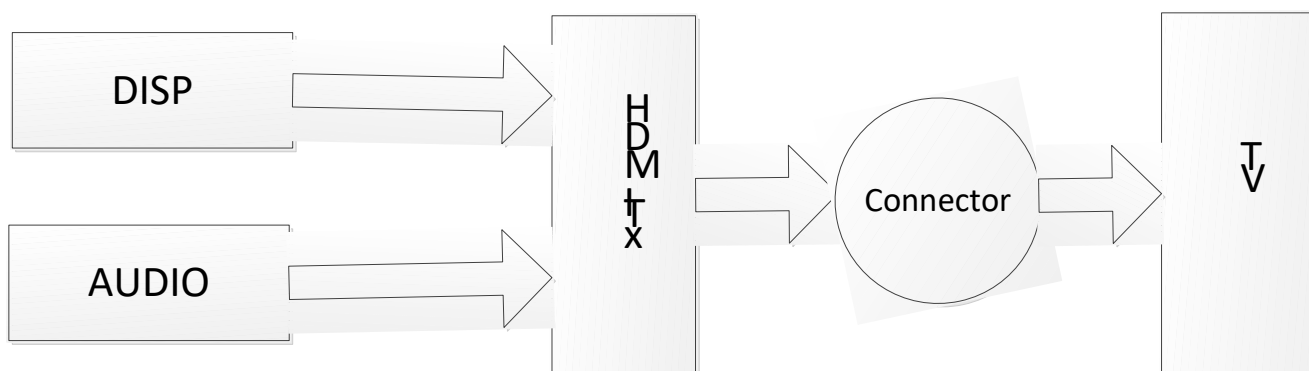
1. SUMMARY

1.1. Module Description

HDMI (High-Definition Multimedia Interface) is a proprietary audio/video interface for transmitting uncompressed video data and compressed or uncompressed digital audio data from an HDMI-compliant source device.

HDMI can be divided into TX and Rx Device. TX is the source device used to transmit data to RX (sink device). At present, this module only supports HDMITX based on HDMI ver1.4.

1.2. Stream Flow



1.3. Platform Support

Platform Features	Taiyaki/Takoyaki	Tiramisu	Muffin	Mochi
1920x1080@24	Y	Y	Y	Y
1920x1080@25	Y	Y	Y	Y
1920x1080@30	Y	Y	Y	Y
1920x1080@50	Y	Y	Y	Y
1920x1080@60	Y	Y	Y	Y
1280x720@60	Y	Y	Y	Y
1280x720@50	Y	Y	Y	Y
720x576@50	Y	Y	Y	Y
720x480@60	Y	Y	Y	Y
1280x1024@60	Y	Y	Y	Y
1024x768@60	Y	Y	Y	Y
1440x900@60	Y	Y	Y	Y
1600x1200@60	Y	Y	Y	Y
1280x800@60	Y	Y	Y	Y

Platform Features	Taiyaki/Takoyaki	Tiramisu	Muffin	Mochi
1366x768@60	Y	Y	Y	Y
1680x1050@60	Y	Y	Y	Y
3840x2160@30	Y	Y	Y	Y
2560x1440@30	Y	Y	Y	Y
Audio_32K	Y	Y	Y	Y
Audio_48K	Y	Y	Y	Y
1920x1080@59.94	N	Y	Y	Y
1920x1080@29.97	N	Y	Y	Y
1280x720@59.94	N	Y	Y	Y
1280x720@29.97	N	Y	Y	Y
3840x2160@29.97	N	Y	Y	Y
800x600@60	N	Y	Y	Y
1280x720@24	N	N	Y	Y
1280x720@25	N	N	Y	Y
1280x720@30	N	N	Y	Y
1920x1080@23.98	N	N	Y	Y
640x480@60	N	N	Y	Y
848x480@60	N	N	Y	Y
1280x768@60	N	N	Y	Y
1280x960@60	N	N	Y	Y
1360x768@60	N	N	Y	Y
1400x1050@60	N	N	Y	Y
1600x900@60	N	N	Y	Y
1920x1200@60	N	N	Y	Y
1920x1440@60	N	N	Y	Y
2560x1600@60	N	N	Y	Y

1.4. Keywords

- EDID: Extended display identification data, which is a metadata format for display devices to describe their capabilities to a video source.
- InfoFrame: Infoframe is a subject of HDMI, used to show information of transmit data.
- DeepColor: A color mode to show mode colorful video.
- SinkInfo: The info from sink device, such as supported timing.

2. API LIST

The MI HDMI module provides the following APIs:

Name of API	Function
MI_HDMI_Init	Initialize HDMI
MI_HDMI_DeInit	Deinitialize HDMI
MI_HDMI_Open	Open HDMI
MI_HDMI_Close	Close HDMI
MI_HDMI_SetAttr	Set HDMI attribute
MI_HDMI_GetAttr	Get HDMI attribute
MI_HDMI_Start	Start HDMI
MI_HDMI_Stop	Stop HDMI
MI_HDMI_GetSinkInfo	Get HDMI Sink side capability information
MI_HDMI_SetAvMute	Set HDMI audio/video mute function
MI_HDMI_ForceGetEdid	Get HDMI EDID information
MI_HDMI_SetDeepColor	Set HDMI Deep Color mode
MI_HDMI_GetDeepColor	Get HDMI Deep Color mode
MI_HDMI_SetInfoFrame	Set HDMI Info Frame
MI_HDMI_GetInfoFrame	Get current HDMI Info Frame
MI_HDMI_SetAnalogDrvCurrent	Set HDMI Channel Driving Current
MI_HDMI_InitDev	Initialize HDMI Device
MI_HDMI_DeInitDev	De-initialize HDMI Device

2.1. MI_HDMI_Init

➤ Function

Initialize HDMI module

➤ Syntax

```
MI_S32 MI_HDMI_Init(MI_HDMI_InitParam_t *pstHdmiPara);
```

➤ Parameter

Parameter Name	Description	Input/Output
pstHdmiParam	Pointer to initialization parameter structure	Input

➤ Return Value

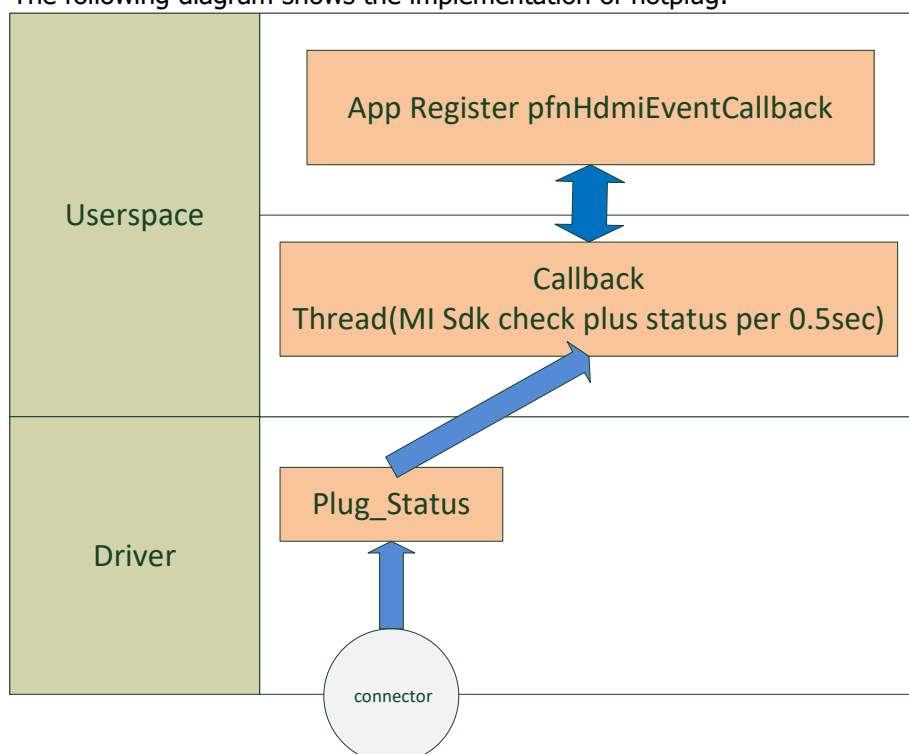
- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

- After HDMI calls back the registered event, system will inform App in the form of callback that a suspected hot-plugging event had happened. If user considers it not necessary to make any special response against the corresponding event, the function pointer in the initialization parameter structure can be set to NULL.
- The following diagram shows the implementation of hotplug:



➤ Example

```

1. MI_HDMI_InitPara_t stHdmiPara;
2. MI_HDMI_Attr_t stAttr;
3. MI_HDMI_TimingType_e eTimingType;
4. /* enable DISP */
5. eTimingType = E_MI_HDMI_TIMING_1080_60P;
6. /* enable HDMI */
7. stHdmiPara.pCallbackArgs = NULL;
8. stHdmiPara.pfnHdmiEventCallback = NULL;
9. MI_HDMI_Init(&stHdmiPara);
10. MI_HDMI_Open(0);
11. /* get support timing of sink device*/
12. MI_HDMI_ForceGetEdid(0, &stEdid);
13. ...
14. /* set the source output timing according to sink edid */
15. MI_MPI_HDMI_GetAttr(0, &stAttr);
16. stAttr.bEnableHdmi = TRUE;
17. stAttr.bEnableVideo = TRUE;
18. stAttr.eVideoFmt = enVideoFmt;
19. stAttr.eVidOutMode= E_MI_HDMI_VIDEO_MODE_YCBCR444;
20. stAttr.eDeepColorMode = E_MI_HDMI_DEEP_COLOR_24BIT;
21. stAttr.bEnableAudio = FALSE;
22. stAttr.bIsMultiChannel = FALSE;
23. stAttr.eBitDepth = E_MI_HDMI_BIT_DEPTH_16;
24. MI_HDMI_SetAttr(0, &stAttr);
25. MI_HDMI_Start(0);
26. /* stop HDMI */
27. MI_HDMI_Stop(0);
28. MI_HDMI_Close(0);
29. MI_HDMI_DeInit();

```

➤ Related API

[MI_HDMI_DeInit](#)

2.2. MI_HDMI_DeInit

➤ Function

Deinitialize HDMI

➤ Syntax

MI_S32 MI_HDMI_DeInit();

➤ Parameter

N/A.

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

N/A.

➤ Example

See the example given in [MI_HDMI_Init](#).

➤ Related API

[MI_HDMI_Init](#)

2.3. MI_HDMI_Open

➤ Function

Open HDMI

➤ Syntax

```
MI_S32 MI_HDMI_Open(MI\_HDMI\_DeviceId\_e eHdmi);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

You should call [MI_HDMI_Init](#) before opening HDMI.
Repeated opening of HDMI will return a successful message.

➤ Example

See the example given in [MI_HDMI_Init](#).

➤ Related API

[MI_HDMI_Close](#)

2.4. MI_HDMI_Close

➤ Function

Close HDMI

➤ Syntax

```
MI_S32 MI_HDMI_Close(MI\_HDMI\_DeviceId\_e eHdmi);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

Repeated closing of HDMI will return a successful message.

➤ Example

See the example given in [MI_HDMI_Init](#).

➤ Related API

[MI_HDMI_Open](#)

2.5. MI_HDMI_SetAttr

➤ Function

Set HDMI attribute

➤ Syntax

```
MI_S32 MI_HDMI_SetAttr(MI\_HDMI\_DeviceId\_e eHdmi, MI\_HDMI\_Attr\_t *pstAttr);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
pstAttr	Pointer to HDMI attribute structure	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h

- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

- You should open HDMI before setting HDMI attribute.
- It is suggested that you set HDMI attribute before calling [MI_HDMI_Start](#).
- If HDMI attribute is to be set after calling MI_HDMI_Start, you should call [MI_HDMI_Stop](#) to disable AV mute and set the new HDMI attribute.
- If the option of audio transfer is enabled, the transfer of video must be enabled too.

➤ Example

```
1. MI_HDMI_InitPara_t stHdmiPara;
2. MI_HDMI_Attr_t stAttr;
3. MI_HDMI_TimingType_e enVideoFmt;
4. /* Enable DISP */
5. /* Set Hdmi attritube */
6. MI_HDMI_Stop(0, MI_TRUE);
7. MI_HDMI_GetAttr(0, &stAttr);
8. /* Change attritube, such as timing */
9. stAttr.eVideoFmt = E_MI_HDMI_VIDEO_FMT_720P_60;
10. MI_HDMI_SetAttr(0, &stAttr);
11. MI_HDMI_Start(0, MI_FALSE);
```

➤ Related API

[MI_HDMI_GetAttr](#)

2.6. MI_HDMI_GetAttr

➤ Function

Get HDMI attribute

➤ Syntax

MI_S32 MI_HDMI_GetAttr([MI_HDMI_DeviceId_e](#) eHdmi, [MI_HDMI_Attr_t](#) *pstAttr);

➤ Syntax

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
pstAttr	Pointer to HDMI attribute structure	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

- Requirement
 - Header: mi_hdmi.h, mi_hdmi_datatype.h
 - Library: libmi_hdmi.a / libmi_hdmi.so

- Note
N/A.

- Example
See the example given in [MI_HDMI_Init](#).

- Related API
[MI_HDMI_SetAttr](#)

2.7. MI_HDMI_Start

- Function
Start HDMI

- Syntax
MI_S32 MI_HDMI_Start([MI_HDMI_DeviceId_e](#) eHdmi);

- Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input

- Return Value
 - Zero: Successful
 - Non-zero: Failed, see [error code for](#) details

- Requirement
 - Header: mi_hdmi.h, mi_hdmi_datatype.h
 - Library: libmi_hdmi.a / libmi_hdmi.so

- Note
Before starting HDMI, call [MI_HDMI_Open](#) first.

- Example
See the example given in [MI_HDMI_Init](#).

- Related API
[MI_HDMI_Stop](#)

2.8. MI_HDMI_Stop

➤ Function

Stop HDMI

➤ Syntax

```
MI_S32 MI_HDMI_Stop(MI\_HDMI\_DeviceId\_e eHdmi);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

N/A.

➤ Related API

[MI_HDMI_Start](#)

2.9. MI_HDMI_GetSinkInfo

➤ Function

Get HDMI Sink side capability information

➤ Syntax

```
MI_S32 MI_HDMI_GetSinkInfo(MI\_HDMI\_DeviceId\_e eHdmi,  
MI\_HDMI\_SinkInfo\_t *pstSinkInfo);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
pstSinkCap	Pointer to HDMI Sink side capability structure	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

- Before calling this function, open HDMI first. To ensure HDMI has been opened successfully, we suggest waiting for 500ms or so after you have called [MI_HDMI_Open](#) interface.
- Call this function after HDMI is started and the cable plugged in.
- This function is normally used when inserting an event handling function or after getting EDID by force.

➤ Example

```
1. MI_HDMI_Init_Para_t stHdmiPara;
2. MI_HDMI_Attr_t stAttr;
3. MI_HDMI_TimingType_e enVideoFmt;
4. MI_HDMI_SinkInfo_t stSinkInfo;
5. /* Enable DISP */
6. /* Enable HDMI */
7. /* Get HDMI Sink Info */
8. MI_HDMI_GetSinkInfo(0, &stSinkInfo);
```

➤ Related API

N/A.

2.10. MI_HDMI_SetAvMute

➤ Function

Set HDMI AV mute function

➤ Syntax

MI_S32 MI_HDMI_SetAvMute([MI_HDMI_DeviceId_e](#) eHdmi, MI_BOOL bAvMute);

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
bAvMute	HDMI AV mute function 1: Enable AV mute 0: Disable AV mute	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

Need to call MI_HDMI_Start before calling this Mute feature.

➤ Example

```
1. MI_HDMI_Start(0);
2. MI_HDMI_SetMute(0, MI_TRUE);
```

➤ Related API

N/A.

2.11. MI_HDMI_ForceGetEdid

➤ Function

Get HDMI EDID information

➤ Syntax

```
MI_S32 MI_HDMI_ForceGetEdid(MI\_HDMI\_DeviceId\_e eHdmi,
MI\_HDMI\_Edid\_t *pstEdidData);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
pstEdidData	HDMI EDID information	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

Before calling this function, open HDMI first.

➤ Example

```
1. MI_HDMI_InitPara_t stHdmiPara;
2. MI_HDMI_Attr_t stAttr;
3. MI_HDMI_Edid_t stEdidData;
4. /* Enable DISP */
5. /* Enable HDMI */
6. /* Get HDMI EDID */
7. MI_HDMI_ForceGetEdid(0, &stEdidData);
```

➤ Related API

N/A.

2.12. MI_HDMI_SetDeepColor

➤ Function

Set Deep Color mode

➤ Function

MI_S32 MI_HDMI_SetDeepColor([MI_HDMI_DeviceId_e](#) eHdmi,
[MI_HDMI_DeepColor_e](#) eDeepColor);

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
eDeepColor	HDMI Deep Color mode	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

- This function is a high-level interface, normally not in use
- Before calling this function, open HDMI first
- This interface can be used to set the Deep Color mode
- **This API is currently not supported.**

➤ Example

```

1. MI_HDMI_InitPara_t stHdmiPara;
2. MI_HDMI_Attr_t stAttr;
3. /* Enable DISP*/
4. /* Enable HDMI */
5. /* Set Deep color mode */
6. MI_HDMI_SetDeepColor (0, E_MI_HDMI_DEEP_COLOR_24BIT);

```

➤ Related API

N/A.

2.13. MI_HDMI_GetDeepColor

➤ Function

Get Deep Color mode

➤ Syntax

```
MI_S32 MI_HDMI_GetDeepColor(MI\_HDMI\_DeviceId\_e eHdmi,
MI\_HDMI\_DeepColor\_e *peDeepColor);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
peDeepColor	Pointer to HDMI Deep Color mode	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

- This function is a high-level interface, normally not in use.
- Before calling this function, open HDMI first.
- **This API is currently not supported.**

➤ Example

N/A.

➤ Related API

[MI_HDMI_SetDeepColor](#)

2.14. MI_HDMI_SetInfoFrame

➤ Function

Set HDMI InfoFrame

➤ Syntax

```
MI_S32 MI_HDMI_SetInfoFrame(MI\_HDMI\_DeviceId\_e eHdmi,  
MI\_HDMI\_InfoFrame\_t *pstInfoFrame);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
pstInfoFrame	Pointer to image InfoFrame structure	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

- This interface is normally used in user state callback function for setting InfoFrame attribute. For other cases, this interface is basically not required.
- Before calling this function, open HDMI first
- This interface supports setting of AVI, SPD and AUDIO InfoFrame only. Setting other than these types of InfoFrame would be invalid.

➤ Example

```
1. MI_HDMI_DeviceId_e eHdmi = E_MI_HDMI_ID_0;
2. MI_HDMI_InfoFrame_t stAviInfoFrame = {E_MI_HDMI_INFOFRAME_TYPE_AVI};
3. MI_HDMI_InfoFrame_t stAudInfoFrame = {E_MI_HDMI_INFOFRAME_TYPE_AUDIO};
4. MI_HDMI_InfoFrame_t stSpdInfoFrame = {E_MI_HDMI_INFOFRAME_TYPE_SPD};
5.
6. /* AviInfoFrame */
7. stAviInfoFrame.unInforUnit.stAviInfoFrame.bEnableAfdOverWrite = 0;
8. ...
9.
10. /* AudioInfoFrame */
11. stAudInfoFrame.unInforUnit.stAudInfoFrame.u32ChannelCount = 2;
12. ...
13.
14. /* SpdInfoFrame */
15. stSpdInfoFrame.unInforUnit.stSpdInfoFrame.au8VendorName[0] = 0;
16. ...
17.
18. MI_HDMI_SetInfoFrame(eHdmi, &stAviInfoFrame);
```

➤ Related API

[MI_HDMI_GetInfoFrame](#)

2.15. MI_HDMI_GetInfoFrame

➤ Function

Get HDMI InfoFrame

➤ Syntax

```
MI_RESULT MI_HDMI_GetInfoFrame(MI\_HDMI\_DeviceId\_e eHdmi,  
MI\_HDMI\_InfoFrameType\_e eInfoFrameType,  
MI\_HDMI\_InfoFrame\_t* pstInfoFrame);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number Parameter range: 0	Input
eInfoFrameType	InfoFrame type	Input
pstInfoFrame	Pointer to image InfoFrame structure	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_hdmi.h, mi_hdmi_datatype.h
- Library: libmi_hdmi.a / libmi_hdmi.so

➤ Note

- This interface is normally used in user state callback function for setting InfoFrame attribute. For other cases, this interface is basically not required.

2.16. MI_HDMI_SetAnalogDrvCurrent

➤ Function

Set HDMI Channel Driving Current

➤ Syntax

```
MI_RESULT MI_HDMI_SetAnalogDrvCurrent(MI\_HDMI\_DeviceId\_e eHdmi,  
MI\_HDMI\_AnalogDrvCurrent\_t *pstAnalogDrvCurrent);
```

➤ Parameter

Parameter Name	Description	Input/Output
eHdmi	HDMI interface number	Input

Parameter Name	Description	Input/Output
	Parameter range: 0	
pstAnalogDrvCurrent	Structure of HDMI Channel Current	Input

- Return Value
 - Zero: Successful
 - Non-zero: Failed, see [error code for](#) details
- Requirement
 - Header: mi_hdmi.h, mi_hdmi_datatype.h
 - Library: libmi_hdmi.a / libmi_hdmi.so
- Note
 - This interface is a high-level interface which allows users to adjust channel current.
 - Before calling this function, open HDMI first.
 - Use before setting is recommended.
- Example


```
1. MI_HDMI_AnalogDrvCurrent_t CurInfo = {};
2.
3. memset(&CurInfo, 0xFF, sizeof(CurInfo));
4. MI_HDMI_SetAnalogDrvCurrent(g_HdmiAttr.eHdmi, &CurInfo);
```

2.17. MI_HDMI_InitDev

- Function

Initialize HDMI Device.
- Syntax


```
MI_S32 MI_HDMI_InitDev(MI\_HDMI\_InitParam\_t *pstInitParam);
```
- Parameter

Parameter Name	Description	Input/Output
pstInitParam	HDMI Init Param	Input
- Return Value
 - Zero: Successful
 - Non-zero: Failed, see [error code for](#) details
- Requirement
 - Header: mi_hdmi.h, mi_hdmi_datatype.h
 - Library: libmi_hdmi.a / libmi_hdmi.so
- Note
 - For Version 2.06 and above, we recommend using this interface to replace the original

[MI_HDMI_Init](#) interface.

- Example
NA.

2.18. MI_HDMI_DeInitDev

- Function
De-initialize HDMI Device.
- Syntax
`MI_S32 MI_HDMI_DeInitDev(void);`
- Parameter
NA.
- Return Value
 - Zero: Successful
 - Non-zero: Failed, see [error code for](#) details
- Requirement
 - Header: `mi_hdmi.h`, `mi_hdmi_datatype.h`
 - Library: `libmi_hdmi.a` / `libmi_hdmi.so`
- Note
 - For Version 2.06 and above, we recommend using this interface to replace the original [MI_HDMI_Deinit](#) interface.
- Example
NA.

3. HDMI DATA TYPE

The HDMI related data types are listed in the table below:

Data type	Description
MI_HDMI_DeviceId_e	Define HDMI interface number
MI_HDMI_InitParam_t	Define HDMI initialization attribute structure
MI_HDMI_EventCallBack	Define HDMI callback function
MI_HDMI_EventType_e	Define HDMI event notification enumeration
MI_HDMI_Attr_t	Define HDMI attribute structure
MI_HDMI_TimingType_e	Define HDMI video format enumeration
MI_HDMI_OutputMode_e	Define HDMI color space type enumeration
MI_HDMI_ColorType_e	Define HDMI color type enumeration
MI_HDMI_DeepColor_e	Define HDMI deep color mode enumeration
MI_HDMI_SampleRate_e	Define HDMI audio output sampling rate enumeration
MI_HDMI_BitDepth_e	Define HDMI audio output sampling bit width enumeration
MI_HDMI_SinkInfo_t	Define HDMI Sink side capability level structure
MI_HDMI_Edid_t	Define HDMI EDID information structure
MI_HDMI_InfoFrameType_e	Define HDMI InfoFrame type enumeration
MI_HDMI_AviInfoFrameVer_t	Define AVI InfoFrame structure
MI_HDMI_AudInfoFrameVer_t	Define AUDIO InfoFrame structure
MI_HDMI_SpdInfoFrame_t	Define SPD InfoFrame structure
MI_HDMI_InfoFrameUnit_u	Define HDMI InfoFrame complex
MI_HDMI_AnalogDrvCurrent_t	Define HDMI Channel Current structure

NOTE: This section includes most of the key data types. For data types not listed here, please refer to `mi_hdmi_datatype.h`.

3.1. MI_HDMI_DeviceId_e

➤ Description

Define HDMI interface number

➤ Definition

```
typedef enum
{
    E_MI_HDMI_ID_0 = 0,
    E_MI_HDMI_ID_BUTT
} MI_HDMI_DeviceId_e;
```

➤ Member

Member	Description
E_MI_HDMI_ID_0	HDMI interface 0

➤ Note

N/A.

➤ Related Data Type and Interface

N/A.

3.2. MI_HDMI_InitParam_t

➤ Description

Define HDMI initialization attribute structure

➤ Definition

```
typedef struct MI_HDMI_InitParam_s
{
    MI\_HDMI\_EventCallback pfnHdmiEventCallback;
    void *pCallBackArgs;
} MI_HDMI_InitParam_t;
```

➤ Member

Member	Description
pfnHdmiEventCallback	Event callback function
pCallBackArgs	Callback function parameter private data

➤ Note

After HDMI initialization, the corresponding hot-plugging event will be returned as long as the callback of Init is not null. If the reporting of such event is not required, you can set the function pointer in the initialization parameter structure to NULL.

➤ Related Data Type and Interface

[MI_HDMI_Init](#)

3.3. MI_HDMI_EventCallBack

➤ Description

Define HDMI callback function

➤ Definition

```
typedef MI_RESULT (*MI_HDMI_EventCallBack)
(MI_HDMI_DeviceId_e eHdmi, MI_HDMI_EventType_e event, void    entParam,
void *pUsrParam);
```

➤ Member

Member	Description
MI_HDMI_EventType_e	HDMI event type
pEventParam pUsrParam	Event private data

➤ Note

N/A.

➤ Related Data Type and Interface

N/A.

3.4. MI_HDMI_EventType_e

➤ Description

Define HDMI event notification enumeration

➤ Definition

```
typedef enum
{
    E_MI_HDMI_EVENT_HOTPLUG = 0,
    E_MI_HDMI_EVENT_NO_PLUG,
    E_MI_HDMI_EVENT_MAX
} MI_HDMI_EventType_e;
```

➤ Member

Member	Description
E_MI_HDMI_EVENT_HOTPLUG	HDMI hot-plugging event
E_MI_HDMI_EVENT_NO_PLUG	HDMI Cable no connection event

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_Init](#)

3.5. MI_HDMI_Attr_t

➤ Description

Define HDMI attribute structure

➤ Definition

```
typedef struct MI_HDMI_Attr_s
{
    MI_HDMI_VideoAttr_t stVideoAttr;
    MI_HDMI_AudioAttr_t stAudioAttr;
    MI_HDMI_EnInfoFrame_t stEnInfoFrame;
} MI_HDMI_Attr_t;

typedef struct MI_HDMI_VideoAttr_s
{
    MI_BOOL                bEnableVideo;
    MI_HDMI_TimingType_e eTimingType;
    MI_HDMI_OutputMode_e eOutputMode;
    MI_HDMI_ColorType_e eColorType;
    MI_HDMI_ColorType_e eInColorType;
    MI_HDMI_ColorType_e eColorType;
    MI_HDMI_DeepColor_e eDeepColorMode;
} MI_HDMI_VideoAttr_t;

typedef struct MI_HDMI_AudioAttr_s
{
    MI_BOOL                bEnableAudio;
    MI_BOOL                bIsMultiChannel;
    MI_HDMI_SampleRate_e eSampleRate;
    MI_HDMI_BitDepth_e eBitDepth;
} MI_HDMI_AudioAttr_t;

typedef struct MI_HDMI_EnInfoFrame_s
{
    MI_BOOL                bEnableAviInfoFrame;
    MI_BOOL                bEnableAudInfoFrame;
    MI_BOOL                bEnableSpdInfoFrame;
    MI_BOOL                bEnableMpegInfoFrame;
} MI_HDMI_EnInfoFrame_t;
```

➤ Member

Member	Description
bEnableVideo	Video enable
eTimingType	Video format. This parameter must be consistent with the video output configuration format.
eOutputMode	HDMI video output mode
eColorType	HDMI output color type
eInColorType	HDMI input color type

Member	Description
eDeepColorMode	DeepColor output mode MI_HDMI_DEEP_COLOR_24BIT is by default not supported. Supports only MI_HDMI_DEEP_COLOR_30BIT, MI_HDMI_DEEP_COLOR_36BIT, and MI_HDMI_DEEP_COLOR_48BIT.
bEnableAudio	Audio enable
bIsMultiChannel	Multi-channel selection 0: 2-channel 1: 8-channel
enSampleRate	Audio sampling rate. This parameter must be consistent with the AO configuration.
enBitDepth	Audio bit width, default is 16. This parameter must be consistent with the AO configuration.
bEnableAviInfoFrame	AVI InfoFrame enable Enable is suggested.
bEnableAudInfoFrame	Audio InfoFrame enable Enable is suggested
bEnableSpdInfoFrame	SPD InfoFrame enable

➤ Note

You may set the HDMI attribute based on the suggested values.

➤ Related Data Type and Interface

[MI_HDMI_SetAttr](#)

3.6. MI_HDMI_TimingType_e

➤ Description

Define HDMI video format enumeration

➤ Definition

```
typedef enum
{
    E_MI_HDMI_TIMING_480_60I    = 0,
    E_MI_HDMI_TIMING_480_60P    = 1,
    E_MI_HDMI_TIMING_576_50I    = 2,
    E_MI_HDMI_TIMING_576_50P    = 3,
    E_MI_HDMI_TIMING_720_50P    = 4,
    E_MI_HDMI_TIMING_720_60P    = 5,
    E_MI_HDMI_TIMING_1080_50I   = 6,
    E_MI_HDMI_TIMING_1080_50P   = 7,
    E_MI_HDMI_TIMING_1080_60I   = 8,
    E_MI_HDMI_TIMING_1080_60P   = 9,
```

E_MI_HDMI_TIMING_1080_30P = 10,
E_MI_HDMI_TIMING_1080_25P = 11,
E_MI_HDMI_TIMING_1080_24P = 12,
E_MI_HDMI_TIMING_4K2K_30P = 13,
E_MI_HDMI_TIMING_1440_50P = 14,
E_MI_HDMI_TIMING_1440_60P = 15,
E_MI_HDMI_TIMING_1440_24P = 16,
E_MI_HDMI_TIMING_1440_30P = 17,
E_MI_HDMI_TIMING_1470_50P = 18,
E_MI_HDMI_TIMING_1470_60P = 19,
E_MI_HDMI_TIMING_1470_24P = 20,
E_MI_HDMI_TIMING_1470_30P = 21,
E_MI_HDMI_TIMING_1920x2205_24P = 22,
E_MI_HDMI_TIMING_1920x2205_30P = 23,
E_MI_HDMI_TIMING_4K2K_25P = 24,
E_MI_HDMI_TIMING_4K1K_60P = 25,
E_MI_HDMI_TIMING_4K2K_60P = 26,
E_MI_HDMI_TIMING_4K2K_24P = 27,
E_MI_HDMI_TIMING_4K2K_50P = 28,
E_MI_HDMI_TIMING_2205_24P = 29,
E_MI_HDMI_TIMING_4K1K_120P = 30,
E_MI_HDMI_TIMING_4096x2160_24P = 31,
E_MI_HDMI_TIMING_4096x2160_25P = 32,
E_MI_HDMI_TIMING_4096x2160_30P = 33,
E_MI_HDMI_TIMING_4096x2160_50P = 34,
E_MI_HDMI_TIMING_4096x2160_60P = 35,
E_MI_HDMI_TIMING_1024x768_60P = 36,
E_MI_HDMI_TIMING_1280x1024_60P = 37,
E_MI_HDMI_TIMING_1440x900_60P = 38,
E_MI_HDMI_TIMING_1600x1200_60P = 39,
E_MI_HDMI_TIMING_1280x800_60P = 40,
E_MI_HDMI_TIMING_1366x768_60P = 41,
E_MI_HDMI_TIMING_1680x1050_60P = 42,
E_MI_HDMI_TIMING_1920x1080_5994P = 43,
E_MI_HDMI_TIMING_1920x1080_2997P = 44,
E_MI_HDMI_TIMING_1280x720_5994P = 45,
E_MI_HDMI_TIMING_1280x720_2997P = 46,
E_MI_HDMI_TIMING_3840x2160_2997P = 47,
E_MI_HDMI_TIMING_800x600_60P = 48,
E_MI_HDMI_TIMING_1280x720_24P = 49,
E_MI_HDMI_TIMING_1280x720_25P = 50,
E_MI_HDMI_TIMING_1280x720_30P = 51,
E_MI_HDMI_TIMING_1920x1080_2398P = 52,
E_MI_HDMI_TIMING_640x480_60P = 53,
E_MI_HDMI_TIMING_848x480_60P = 54,
E_MI_HDMI_TIMING_1280x768_60P = 55,
E_MI_HDMI_TIMING_1280x960_60P = 56,
E_MI_HDMI_TIMING_1360x768_60P = 57,
E_MI_HDMI_TIMING_1400x1050_60P = 58,
E_MI_HDMI_TIMING_1600x900_60P = 59,
E_MI_HDMI_TIMING_1920x1200_60P = 60,
E_MI_HDMI_TIMING_1920x1440_60P = 61,
E_MI_HDMI_TIMING_2560x1600_60P = 62,
E_MI_HDMI_TIMING_MAX,

```
} MI_HDMI_TimingType_e;
```

➤ Member

N/A.

➤ Note

- It is essential that you set the corresponding HDMI format based on the video output format.
- The timing set by App must match with the DISP module timing. That is, the HDMI timing must be consistent with DISP module timing.

➤ Related Data Type and Interface

[MI_HDMI_SetAttr](#)

3.7. MI_HDMI_OutputMode_e

➤ Description

Define HDMI output mode type enumeration

➤ Definition

```
typedef enum
{
    E_MI_HDMI_OUTPUT_MODE_HDMI = 0,
    E_MI_HDMI_OUTPUT_MODE_HDMI_HDCP,
    E_MI_HDMI_OUTPUT_MODE_DVI,
    E_MI_HDMI_OUTPUT_MODE_DVI_HDCP,
    E_MI_HDMI_OUTPUT_MODE_AUTO,
    E_MI_HDMI_OUTPUT_MODE_MAX,
} MI_HDMI_OutputMode_e;
```

➤ Member

Member	Description
E_MI_HDMI_OUTPUT_MODE_HDMI	HDMI Output Mode
E_MI_HDMI_OUTPUT_MODE_HDMI_HDCP	HDMI with HDCP Output Mode
E_MI_HDMI_OUTPUT_MODE_DVI	DVI Output Mode
E_MI_HDMI_OUTPUT_MODE_DVI_HDCP	DVI with HDCP Output Mode
E_MI_HDMI_OUTPUT_MODE_AUTO	Automatic selection via EDID

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_SetAttr](#)

3.8. MI_HDMI_ColorType_e

➤ Description

Define HDMI color space type enumeration

➤ Definition

```
typedef enum
{
    E_MI_HDMI_COLOR_TYPE_RGB444 = 0,
    E_MI_HDMI_COLOR_TYPE_YCBCR422,
    E_MI_HDMI_COLOR_TYPE_YCBCR444,
    E_MI_HDMI_COLOR_TYPE_YCBCR420,
    E_MI_HDMI_COLOR_TYPE_AUTO,
    E_MI_HDMI_COLOR_TYPE_MAX
} MI_HDMI_ColorType_e;
```

➤ Member

Member	Description
E_MI_HDMI_COLOR_TYPE_RGB444	RGB444 Output Mode
E_MI_HDMI_COLOR_TYPE_YCBCR422	YCBCR422 Output Mode
E_MI_HDMI_COLOR_TYPE_YCBCR444	YCBCR444 Output Mode
E_MI_HDMI_COLOR_TYPE_YCBCR420	YCBCR420 Output Mode
E_MI_HDMI_COLOR_TYPE_AUTO	Automatic selection via EDID

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_SetAttr](#)

3.9. MI_HDMI_DeepColor_e

➤ Description

Define HDMI deep color mode enumeration

➤ Definition

```
typedef enum
{
    E_MI_HDMI_DEEP_COLOR_24BIT = 0x00,
    E_MI_HDMI_DEEP_COLOR_30BIT,
    E_MI_HDMI_DEEP_COLOR_36BIT,
    E_MI_HDMI_DEEP_COLOR_48BIT,
    E_MI_HDMI_DEEP_COLOR_MAX,
} MI_HDMI_DeepColor_e;
```

➤ Member

Member	Description
E_MI_HDMI_DEEP_COLOR_24BIT	HDMI Deep Color 24-bit Mode
E_MI_HDMI_DEEP_COLOR_30BIT	HDMI Deep Color 30-bit Mode
E_MI_HDMI_DEEP_COLOR_36BIT	HDMI Deep Color 36-bit Mode
E_MI_HDMI_DEEP_COLOR_48BIT	HDMI Deep Color 48-bit Mode

➤ Note

N/A.

- Related Data Type and Interface
[MI_HDMI_SetAttr](#)

3.10. MI_HDMI_SampleRate_e

- Description
Define HDMI audio output sampling rate enumeration
- Definition


```
typedef enum
{
    E_MI_HDMI_AUDIO_SAMPLERATE_UNKNOWN    = 0,
    E_MI_HDMI_AUDIO_SAMPLERATE_32K       = 1,
    E_MI_HDMI_AUDIO_SAMPLERATE_44K       = 2,
    E_MI_HDMI_AUDIO_SAMPLERATE_48K       = 3,
    E_MI_HDMI_AUDIO_SAMPLERATE_88K       = 4,
    E_MI_HDMI_AUDIO_SAMPLERATE_96K       = 5,
    E_MI_HDMI_AUDIO_SAMPLERATE_176K      = 6,
    E_MI_HDMI_AUDIO_SAMPLERATE_192K      = 7,
    E_MI_HDMI_AUDIO_SAMPLERATE_MAX,
} MI_HDMI_SampleRate_e;
```
- Member
N/A.
- Note
When the App is setting the sampling rate, it needs to match the sampling rate of the Audio module, and the sampling rate of HDMI needs to be consistent with the sampling rate of the Audio module.
- Related Data Type and Interface
[MI_HDMI_SetAttr](#)

3.11. MI_HDMI_BitDepth_e

- Description
Define HDMI audio output sampling bit width enumeration
- Definition


```
typedef enum
{
    E_MI_HDMI_BIT_DEPTH_UNKNOWN = 0,
    E_MI_HDMI_BIT_DEPTH_8      = 8,
    E_MI_HDMI_BIT_DEPTH_16     = 16,
    E_MI_HDMI_BIT_DEPTH_18     = 18,
    E_MI_HDMI_BIT_DEPTH_20     = 20,
    E_MI_HDMI_BIT_DEPTH_24     = 24,
    E_MI_HDMI_BIT_DEPTH_32     = 32,
    E_MI_HDMI_BIT_DEPTH_BUTT
}
```



```
}MI_HDMI_BitDepth_e;
```

➤ Member

N/A.

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_SetAttr](#)

3.12. MI_HDMI_SinkInfo_t

➤ Description

Define HDMI Sink side capability structure

➤ Definition

```
typedef struct MI_HDMI_Sink_Info_s
{
    MI_BOOL bConnected;
    MI_BOOL bSupportHdmi;
    MI_HDMI_TimingType_e eNativeTimingType;
    MI_BOOL abVideoFmtSupported[E_MI_HDMI_TIMING_MAX];
    MI_BOOL bSupportYCbCr444;
    MI_BOOL bSupportYCbCr422;
    MI_BOOL bSupportYCbCr;
    MI_BOOL bSupportxvYcc601;
    MI_BOOL bSupportxvYcc709;
    MI_U8 u8MdB;
    MI_BOOL abAudioFmtSupported[MI_HDMI_MAX_ACAP_CNT];
    MI_U32 au32AudioSampleRateSupported[MI_HDMI_MAX_AUDIO_SAMPLE_RATE_CNT];
    MI_U32 u32MaxPcmChannels;
    MI_U8 u8Speaker;
    MI_U8 au8IdManufactureName[4];
    MI_U32 u32IdProductCode;
    MI_U32 u32IdSerialNumber;
    MI_U32 u32WeekOfManufacture;
    MI_U32 u32YearOfManufacture;
    MI_U8 u8Version;
    MI_U8 u8Revision;
    MI_U8 u8EdidExternBlockNum;
    MI_U8 au8IeeRegId[3]; //IEEE registration identifier
    MI_U8 u8PhyAddr_A;
    MI_U8 u8PhyAddr_B;
    MI_U8 u8PhyAddr_C;
    MI_U8 u8PhyAddr_D;
    MI_BOOL bSupportDviDual;
    MI_BOOL bSupportDeepColorYcbcr444;
    MI_BOOL bSupportDeepColor30Bit;
    MI_BOOL bSupportDeepColor36Bit;
    MI_BOOL bSupportDeepColor48Bit;
```

```

MI_BOOL bSupportAi;
MI_U32 u8MaxTmdsClock;
MI_BOOL bILatencyFieldsPresent;
MI_BOOL bLatencyFieldsPresent;
MI_BOOL bHdmiVideoPresent;
MI_U8 u8VideoLatency;
MI_U8 u8AudioLatency;
MI_U8 u8InterlacedVideoLatency;
MI_U8 u8InterlacedAudioLatency;
} MI_HDMI_SinkInfo_t;

```

➤ Member

Member	Description
bConnected	Device connected or not
bSupportHdmi	HDMI supported or not. If HDMI is not supported, the device is a DVI device.
eNativeVideoFormat	Display device physical resolution
abVideoFmtSupported	Video capability 1: The display format is supported 0: The display format is not supported
bSupportYCbCr	YCBCR display supported or not 1: YCBCR display is supported 0: Only RGB display is supported
bSupportxvYCC601	xvYCC601 color format supported or not
bSupportxvYCC709	xvYCC709 color format supported or not
u8MDBit	Transmit profile supported by xvYCC601 1: P0; 2: P1; 4: P2
bAudioFmtSupported	Audio capability. Please refer to EIA-CEA-861-D Table 37 1: Display format supported 0: Display format not supported
u32AudioSampleRateSupported	Audio sampling rate capability. 0 means invalid setting, and others supported audio sampling rate.
u32MaxPcmChannels	Maximum audio PCM channel number
u8Speaker	Speaker position, please refer to SpeakerATABlock definition given in EIA-CEA-861-D.
u8IDManufactureName	Device manufacturer ID
u32IDProductCode	Device ID
u32IDSerialNumber	Device serial number
u32WeekOfManufacture	Device week of manufacture
u32YearOfManufacture	Device year of manufacture
u8Version	Device version
u8Revision	Device revision
au8EDIDExternBlockNum	EDID external block number

Member	Description
au8IEERegId	24-bit IEEE Registration Identifier (0x000C03)
u8PhyAddr_A	CEC physical address A
u8PhyAddr_B	CEC physical address B
u8PhyAddr_C	CEC physical address C
u8PhyAddr_D	CEC physical address D
bSupportDVIDual	DVI dual-link operation supported or not
bSupportDeepColorYCBCR444	YCBCR 4:4:4 Deep Color Mode supported or not
bSupportDeepColor36Bit	Deep Color 36-bit Mode supported or not
bSupportDeepColor48Bit	Deep Color 48-bit Mode supported or not
bSupportAI	AI Mode supported or not
u8MaxTMDSClock	Maximum TMDS clock
bLatencyFieldsPresent	Latency field
bHDMIVideoPresent	Video Latency and Audio Latency fields present or not
u8VideoLatency	Video latency
u8AudioLatency	Audio latency
u8InterlacedVideoLatency	Video latency under interlaced video mode
u8InterlacedAudioLatency	Audio latency under interlaced audio mode

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_GetSinkInfo](#)

3.13. MI_HDMI_Edid_t

➤ Description

Define HDMI EDID information structure

➤ Definition

```
typedef struct MI_HDMI_Edid_s
{
    MI_BOOL bEdidValid;
    MI_U32 u32Edidlength;
    MI_U8 au8Edid[512];
} MI_HDMI_Edid_t;
```

➤ Member

Member	Description
bEdidValid	EDID information valid or not
u32Edidlength	EDID information data length, normally 256 bytes
au8Edid	au8Edid

➤ Note

N/A.

- Related Data Type and Interface

[MI_HDMI_ForceGetEdid](#)

3.14. MI_HDMI_InfoFrameType_e

- Description

Define HDMI InfoFrame type enumeration

- Definition

```
typedef enum
{
    E_MI_INFOFRAME_TYPE_AVI = 0,
    E_MI_INFOFRAME_TYPE_SPD,
    E_MI_INFOFRAME_TYPE_AUDIO,
    E_MI_INFOFRAME_TYPE_MAX
} MI_HDMI_InfoFrameType_e;
```

- Member

Member	Description
E_MI_INFOFRAME_TYPE_AVI	AVI InfoFrame type
E_MI_INFOFRAME_TYPE_SPD	SPD InfoFrame type
E_MI_INFOFRAME_TYPE_AUDIO	AUDIO InfoFrame type

- Note

N/A.

- Related Data Type and Interface

[MI_HDMI_DeInit](#)

[MI_HDMI_SetInfoFrame](#)

3.15. MI_HDMI_AviInfoFrameVer_t

- Description

Define AVI InfoFrame structure

- Definition

```
typedef struct MI_HDMI_AviInfoFrameVer_s
{
    MI_BOOL    enableAFDoverWrite;
    MI_U8    A0Value;
    MI_HDMI_ColorType_e eColorType;
    MI_HDMI_Colorimetry_e eColorimetry;
    MI_HDMI_ExtColorimetry_e eExtColorimetry;
    MI_HDMI_YccQuantRange_e eYccQuantRange;
    MI_HDMI_TimingType_e eTimingType; //trans to MS_VIDEO_TIMING in impl
    MI_HDMI_VideoAfdRatio_e eAfdRatio;
    MI_HDMI_ScanInfo_e eScanInfo;
```

```
MI_HDMI_AspectRatio_e eAspectRatio;  
} MI_HDMI_AviInfoFrameVer_t;
```

➤ Member

Member	Description
eColorType	Color space type
eScanInfo	Scan information enumeration variable
eColorimetry	Color space matrix
eAspectRatio	Aspect ratio enumeration variable
eExtColorimetry	External color space matrix
eTimingType	Video format type
eYccQuantRange	YCC color space range

➤ Note

For the enumeration members of each member variable, see `mi_hdmi_datatype.h`.

➤ Related Data Type and Interface

[MI_HDMI_SetInfoFrame](#)

[MI_HDMI_GetInfoFrame](#)

3.16. MI_HDMI_AudInfoFrameVer_t

➤ Description

Define AUDIO InfoFrame structure

➤ Definition

```
typedef struct MI_HDMI_AudInfoFrameVer_s  
{  
    MI_U32 u32ChannelCount;  
    MI_HDMI_AudioCodeType_e eAudioCodeType;  
    MI_HDMI_SampleRate_e eSampleRate;  
} MI_HDMI_AudInfoFrameVer_t;
```

➤ Member

Member	Description
u32ChannelCount	Channel number
eAudioCodeType	Audio code type
eSampleRate	Sampling rate, suggested value is 0.
	0: Decided by stream header
	1: 32kHz
	2: 44.1 kHz
	3: 48 kHz
	4: 88.2 kHz
	5: 96 kHz
	6: 176.4 kHz

Member	Description
	7: 192 kHz

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_SetInfoFrame](#)[MI_HDMI_GetInfoFrame](#)

3.17. MI_HDMI_SpdInfoFrame_t

➤ Description

Define SPD InfoFrame structure

➤ Definition

```
typedef struct MI_HDMI_SpdInfoFrame_s
{
    MI_U8 au8VendorName[8];
    MI_U8 au8ProductDescription[16];
} MI_HDMI_SpdInfoFrame_t;
```

➤ Member

Member	Description
au8VendorName[8]	Vendor name
au8ProductDescription[16]	Product Description

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_SetInfoFrame](#)[MI_HDMI_GetInfoFrame](#)

3.18. MI_HDMI_InfoFrame_Unit_u

➤ Description

Define HDMI InfoFrame complex

➤ Definition

```
typedef union
{
    MI\_HDMI\_AviInfoFrameVer\_t    stAviInfoFrame;
    MI\_HDMI\_AudInfoFrameVer\_t    stAudInfoFrame;
    MI_HDMI_SpdInfoFrame_t      stSpdInfoFrame;
```

```
} MI_HDMI_InfoFrameUnit_u;
```

➤ Member

Member	Description
stAviInfoFrame	AVI InfoFrame
stAudInfoFrame	AUD InfoFrame
stSpdInfoFrame	SPD InfoFrame

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_SetInfoFrame](#)

[MI_HDMI_GetInfoFrame](#)

3.19. MI_HDMI_AnalogDrvCurrent_t

➤ Description

Define HDMI Channel Driving Current Structure

➤ Definition

```
typedef struct MI_HDMI_AnalogDrvCurrent_s
{
    MI_U8 u8DrvCurTap1Ch0;
    MI_U8 u8DrvCurTap1Ch1;
    MI_U8 u8DrvCurTap1Ch2;
    MI_U8 u8DrvCurTap1Ch3;
    MI_U8 u8DrvCurTap2Ch0;
    MI_U8 u8DrvCurTap2Ch1;
    MI_U8 u8DrvCurTap2Ch2;
    MI_U8 u8DrvCurTap2Ch3;
} MI_HDMI_AnalogDrvCurrent_t;
```

➤ Member

Member	Description
u8DrvCurTap1Ch0	Tap1 Channel 0 Driving Current Setting
u8DrvCurTap1Ch1	Tap1 Channel 1 Driving Current Setting
u8DrvCurTap1Ch2	Tap1 Channel 2 Driving Current Setting
u8DrvCurTap1Ch3	Tap1 Channel 3 Driving Current Setting
u8DrvCurTap2Ch0	Tap2 Channel 0 Driving Current Setting
u8DrvCurTap2Ch1	Tap2 Channel 1 Driving Current Setting

u8DrvCurTap2Ch2	Tap2 Channel 2 Driving Current Setting
u8DrvCurTap2Ch3	Tap2 Channel 3 Driving Current Setting

➤ Note

N/A.

➤ Related Data Type and Interface

[MI_HDMI_SetAnalogDrvCurrent](#)

4. HDMI ERROR CODE

The table below lists the HDMI API error codes.

Error Code	Macro Definition	Description
0xA00B2011	MI_ERR_HDMI_INVALID_PARAM	Invalid parameter
0xA00B2001	MI_ERR_HDMI_INVALID_DEVID	Invalid device ID
0xA00B2018	MI_ERR_HDMI_DRV_FAILED	HDMI drive failed
0xA00B2015	MI_ERR_HDMI_NOT_INIT	HDMI not initialized
0xA00B2008	MI_ERR_HDMI_NOT_SUPPORT	Operation not supported
0xA00B21E0	MI_ERR_HDMI_UNSupport_TIMING	Unsupported output timing
0xA00B21E1	MI_ERR_HDMI_UNSupport_COLORTYPE	Unsupported color type
0xA00B21E2	MI_ERR_HDMI_UNSupport_ACODETYPE	Unsupported audio code type
0xA00B21E3	MI_ERR_HDMI_UNSupport_AFREQ	Unsupported audio frequency
0xA00B21E4	MI_ERR_HDMI_EDID_HEADER_ERR	EDID header decoding failed
0xA00B21E5	MI_ERR_HDMI_EDID_DATA_ERR	EDID data error

5. PROCFS 介绍

5.1. cat

➤ Debug Info

```
# cat /proc/mi_modules/mi_hdmi/mi_hdmi0
```

```
----- HDMI0 Dev Info -----
InitFlag AvMute PowerOn
  Y    N    Y
EnableVideo TimingType OutputMode ColorType DeepColorMode
  1      9      0      2      4
EnableAudio IsMultiChannel BitDepth CodeType SampleRate
  1        0       16      0      3
```

➤ Debug Info Analysis

Record HDMI current usage status and device attribute which can be dynamically got to debug and test.

➤ Parameter Description

Parameter		Description
device info	InitFlag	Whether the HDMI module is initialized Y: Initialized N: Uninitialized
	AvMute	Whether the audio and video are in MUTE state Y: MUTE N: Not MUTE
	PowerOn	Whether HDMI is enabled for power management Y: Yes N: No
	EnableVideo	Whether to enable video 0: Disable 1: Enable
	TimingType	Current HDMI output resolution Value range[0~ E_MI_HDMI_TIMING_MAX]
	OutputMode	HDMI output mode 0: HDMI mode 2: DVI mode 1/3: HDCP mode, not support at present
	ColorType	Color space 0: RGB444

Parameter		Description
		1: YUV422 2: YUV444 3: YUV420
	DeepColorMode	Color bit depth 0: 8Bit 1: 10Bit 2: 12Bit 3: 16Bit
	EnableAudio	Whether to enable audio 0: Disable 1: Enable
	IsMultiChannel	Whether is multi-channel 0: No 1: Yes
	BitDepth	Audio sample bit depth 8: 8Bit 16: 16Bit 32: 32Bit
	CodeType	Compression format of audio output 0: PCM 1: Not PCM
	SampleRate	Audio output sample rate 0: unknown 1: 32K 2: 44K 3: 48K 4: 88K 5: 96K 6: 176K 7: 192K